THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET B

Chapter 12: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In quadrilateral ABCD, the sides opposite to AB and the angle opposite to ∠A are respectively: 1

(a) AD and ∠D **(b)** CD and ∠C **(c)** CD and ∠B **(d)** BC and ∠B

2. Which of the following is NOT a test for a parallelogram? 1

(a) one pair of opposite sides equal
(b) both pairs of opposite sides equal
(c) one pair of opposite sides equal and parallel
(d) diagonals bisect each other

3. The Midpoint Theorem states that the segment joining the midpoints of two sides of a triangle is: 1

(a) parallel to the third side and half of it
(b) perpendicular to the third side
(c) double the third side
(d) equal to the third side

4. The quadrilateral formed by joining the midpoints of the sides of any quadrilateral is a: 1

(a) square **(b)** rhombus **(c)** rectangle **(d)** parallelogram

5. **Assertion (A):** If the diagonals of quadrilateral PQRS bisect each other, PQRS is a parallelogram. 1

Reason (R): The two pairs of triangles formed by the diagonals are congruent by SAS, giving parallel opposite sides.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

6. In a parallelogram ABCD, ∠A = 65°. Then ∠B is: 1

(a) 295° **(b)** 115° **(c)** 65° **(d)** 58°

7. In △ABC, P and Q are the midpoints of AB and AC and BC = 18 cm. The length of PQ is: 1

(a) 9 cm **(b)** 11 cm **(c)** 18 cm **(d)** 36 cm

8. P and Q are the midpoints of sides AB and AC of △ABC and PQ = 7 cm. Then BC is: 1

(a) 3.5 cm **(b)** 7 cm **(c)** 14 cm **(d)** 9 cm

9. G is the centroid of $\triangle ABC$ and GD = 4 cm, where AD is a median. Then AD is: 1

(a) 16 cm **(b)** 8 cm **(c)** 12 cm **(d)** 13 cm

10. The diagonals of a quadrilateral ABCD are AC = 10 cm and BD = 14 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is: 1

(a) 24 cm **(b)** 48 cm **(c)** 70 cm **(d)** 12 cm

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State the 'equal parallel sides' test for a parallelogram. 2

12. D and E are the midpoints of sides AB and AC of $\triangle ABC$ and $DE = 7.5$ cm. Find BC and state the relation between DE and BC. 2

13. G is the centroid of $\triangle ABC$ and AD is a median of length 21 cm. Find AG and GD. 2

14. The diagonals of a quadrilateral are 9 cm and 13 cm. Find the perimeter of the parallelogram formed by joining the midpoints of its sides. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The diagonal AC of parallelogram ABCD bisects $\angle A$. Show that ABCD is a rhombus. 3

16. State and prove the Midpoint Theorem (outline with the auxiliary line). 3

17. Prove: the line through the midpoint of one side of a triangle, parallel to another side, bisects the third side. 3

OR

Prove that the midpoints of the sides of any quadrilateral form a parallelogram. 3

18. In a parallelogram ABCD, $\angle A = (3x + 15)^\circ$ and $\angle B = (2x + 5)^\circ$. Find x and all four angles. 3

OR

The angles of a quadrilateral are in the ratio 2 : 3 : 3 : 4. Find each angle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. State and prove the converse of the Midpoint Theorem. In $\triangle ABC$, P is the midpoint of AB and $PQ \parallel BC$ with Q on AC; if BC $= 30$ cm and AC $= 22$ cm, find PQ and AQ. 5

OR

Prove the Centroid Theorem: the medians of a triangle are concurrent and divide each other in the ratio 2 : 1. (Use midpoints of CM and BM.) 5

20. Prove that the midpoints of the sides of a quadrilateral ABCD form a parallelogram. If the diagonals are AC $= 16$ cm and BD $= 16$ cm, name the parallelogram and find its perimeter. 5

OR

In quadrilateral ABCD, P, Q, R, S are the midpoints of AB, BC, CD, DA. Show that PR and QS bisect each other and that if AC $=$ BD then PR $\perp$ QS. 5

END OF QUESTION PAPER — **Total: 40 Marks**